

STANDARD OPERATING PROCEDURE

Inspection of Precision Shaft Assembly - PS-2024-001

SOP No: SOP-INS-2024-031

Rev: B

Effective: 2024-11-01

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1. PURPOSE

This procedure defines the inspection steps for Precision Shaft Assembly Part #PS-2024-001. All critical dimensions must be verified against engineering drawing ED-PS-2024-001 Rev A.

2. EQUIPMENT REQUIRED

- Outside Micrometer (0-25mm range, 0.001mm resolution)
- Bore Gauge (10mm, 0.001mm resolution)
- Height Gauge (0-300mm, 0.01mm resolution)
- Thread Gauge (M12 pitch diameter)
- Profilometer (Ra measurement capability)
- Optical comparator (optional, for thread profile verification)

3. SETUP INSTRUCTIONS

- Verify all measurement instruments have current calibration certificates.
- Clean the workpiece with isopropyl alcohol before inspection.
- Allow the part to stabilize at $20^{\circ}\text{C} \pm 2^{\circ}\text{C}$ for a minimum of 30 minutes.
- Set up the micrometer with zero verification using a calibration standard.
- Prepare the inspection report form (Form QC-INS-001).

4. INSPECTION STEPS

Step 1: Visual Inspection

Examine all surfaces for tool marks, burrs, and surface defects.
Accept/Reject per visual standard VS-001.

Step 2: Outer Diameter Measurement

Measure OD at three positions (25%, 50%, 75% of length) using micrometer.
Specification: $\varnothing 25.400 \pm 0.013\text{mm}$. Record all three readings.

Step 3: Bore Diameter Measurement

Measure bore ID using bore gauge at two positions.
Specification: $\varnothing 10.000 +0.018/-0.000\text{mm}$. Record readings.

Step 4: Overall Length Verification

Measure overall length using height gauge on surface plate.
Specification: $150.000 \pm 0.050\text{mm}$.

Step 5: Thread Pitch Diameter Check

Verify thread pitch diameter using thread gauge.
Specification: $M12.700 \pm 0.025\text{mm}$. Go/No-Go check required.

Step 6: Surface Finish Verification

Measure surface roughness using profilometer at three locations.
Specification: $Ra 0.800$ (max 1.600) μm .

Step 7: Final Documentation

Complete inspection report. Verify all dimensions recorded.